THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** symptom	Felt by the patient
2	**(b)** Diabetes	Lifestyle related
3	**(a)** chronic diseases	Long lasting
4	**(b)** roundworms	Infects intestine
5	**(c)** avoiding mosquito breeding in still water	Control the vector
6	**(d)** magnet	An electromagnet
7	**(b)** reverse the direction of current	Polarity depends on current direction
8	**(a)** movement of liquid iron in the core	Creates electric currents
9	**(b)** nichrome	Used in heating elements
10	**(c)** the current increases	More current gives more heat

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bacteria that were killed earlier survive an antibiotic (1). They multiply despite the treatment (1).	1+1
12	Use antibiotics only when prescribed (1). Take the correct dose for the full duration (1).	1+1
13	A cell with two different metal electrodes in an electrolyte (1). Chemical reaction produces electricity (1).	1+1
14	Zinc container (negative terminal) (1). Carbon rod with metal cap (positive) in paste electrolyte (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Balanced diet supplies nutrients (1). Exercise and sleep strengthen body (1). Avoiding tobacco and alcohol reduces risk (1).	1+1+1
16	Dry cell is single-use (1). Rechargeable can be reused many times (1). Rechargeable wears out after many cycles (1).	1+1+1
17	Fleming studied harmful bacteria in 1928 (1). A mould on a discarded dish stopped the bacteria (1). Mould substance became penicillin, the first antibiotic (1).	1+1+1
	OR: Rest helps the body recover (1½). It limits spread of the pathogens to others (1½).	1½+1½
18	It still contains acids and metals (1). They can cause fires (1). They harm the environment, but can be recycled (1).	1+1+1
	OR: Zinc container is the negative terminal (1½). Carbon rod with metal cap is the positive terminal (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ayurveda: balance of body, mind and surroundings (1½). Dinacharya and ritucharya routines (1). Variolation against smallpox (1). Modern vaccines are safer and tested (1½).	1½+1+1+1½
	OR: Mould on a dish stopped bacteria (1). Penicillin was the first antibiotic (1). Antibiotics treat bacterial infections (1). Vaccines prevent diseases by training immunity (1). Antibiotics do not work on viruses (1).	1+1+1+1+1
20	(i) Coil with iron nail; nichrome coil (2). (ii) Magnetic effect and heating effect (1). (iii) Insulated wire for both; do not touch hot coil (2).	2+1+2
	OR: Liquid iron moves in the core (1). It creates currents and a magnetic field (1). Animals navigate with it (1). Field blocks harmful space particles (1). Protects life on Earth (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** preventive	Used before the disease
2	**(a)** bacterial infections	They target bacterial cells
3	**(d)** take antibiotics only when prescribed	Complete the right dose
4	**(a)** Regular exercise and adequate sleep	Healthy lifestyle
5	**(c)** vaccination and isolation	Viral diseases
6	**(b)** electrodes	Dipped in electrolyte
7	**(a)** lemon juice	Acidic liquid
8	**(d)** zinc container	Container acts as negative
9	**(c)** mobile phones and laptops	Reusable
10	**(b)** heating effect of current	Filament glows

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A scientist who studied bacteria (1). He discovered penicillin from a mould in 1928 (1).	1+1
12	An old Indian method of protecting against smallpox (1). Material from a sore was used to cause mild infection and immunity (1).	1+1
13	Chemicals get used up (1). It cannot be recharged and must be disposed of (1).	1+1
14	Mobile phones and laptops (1). Inverters or electric vehicles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A practice of scratching skin with smallpox material to cause a mild infection (1½). Teekedaars were the people who performed it (1½).	1½+1½
16	Two cells give more current (1). More current means more heat in filament (1). Hotter filament glows brighter (1).	1+1+1
17	Water – boil it (1). Food – cook and cover it (1). Personal items – do not share towels (1).	1+1+1
	OR: She was a biomedical researcher (1). She studied links of hormones and viruses with cancer (1). Her work showed risks from tobacco and pollution (1).	1+1+1
18	Moving liquid iron in the core forms currents (1). Currents produce Earth's magnetic field (1). Animals use it to navigate (1).	1+1+1
	OR: It becomes very hot (1½). It can cause burns (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Typhoid by contaminated food and water, so boil water and keep sanitation (2). Dengue by mosquitoes, so avoid stagnant water and use nets (2). Community drives and awareness (1).	2+2+1
	OR: Smallpox had a single stable virus and effective vaccine (1½). Mass vaccination stopped its spread (1). Influenza viruses change often (1). New vaccines are needed (1½).	1½+1+1+1½
20	LED has two leads (1). Longer lead is positive (1). Connect it to the positive terminal of battery (1). Shorter lead to negative (1). If it does not glow, reverse connections (1).	1+1+1+1+1
	OR: Too many appliances draw large current (1½). Wires heat up (1). Plastic parts may melt (1). It may start a fire (1½).	1½+1+1+1½

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ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** sign	Observed or measured
2	**(a)** vitamin C	A deficiency disease
3	**(d)** protozoan	Spread by mosquitoes
4	**(c)** vector	e.g. mosquito, housefly
5	**(d)** removing pathogens	Hygiene
6	**(b)** an iron core	Iron strengthens the field
7	**(a)** the current is switched off	It is a temporary magnet
8	**(a)** navigate	They sense the field
9	**(d)** nichrome	High resistance alloy
10	**(d)** current flows for a longer time	More time gives more heat

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diabetes (1). Cancer or asthma (1).	1+1
12	Mosquito nets or repellents (1). Avoid water stagnation to stop breeding (1).	1+1
13	They contain harmful metals like lead (1). Recycling prevents pollution and recovers materials (1).	1+1
14	Galvani thought electricity came from the frog (1). Volta showed it came from metals and liquid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It can make bacteria in animals resistant (1). Resistance spreads via food and manure to humans (1). Infections become hard to treat (1).	1+1+1
16	Magnetic: current makes a coil act as a magnet (1). Use: crane (½). Heating: wire gets hot (1). Use: heater (½).	1+½+1+½
17	Vaccination controlled smallpox and polio (1½). Lifestyle changes increased diabetes (1½).	1½+1½
	OR: Ayurveda, Siddha, Unani (1). Benefit: natural substances and healthy lifestyle (1). Limitation: may not work for all diseases and stages (1).	1+1+1
18	Bring compass near an end (1). Needle's north-seeking end points to the south pole of the coil (1). Opposite end is the other pole (1).	1+1+1
	OR: Use wires of suitable thickness (1½). Use plugs and sockets rated for the current (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	TB: bacteria affecting lungs, cough, fever (1½). Prevention: avoid close contact, mask, vaccination (1). Hepatitis A: virus affecting liver, jaundice (1½). Prevention: boiled water, vaccination (1).	1½+1+1½+1
	OR: Hygiene prevents contact spread (1). Clean water prevents cholera and typhoid (1½). Sanitation prevents worms like Ascaris (1½). Handwashing and cooked food (1).	1+1½+1½+1
20	Electromagnet is temporary (1). Strength can be changed (1). Poles can be reversed (1). Can be switched on and off (1). Permanent magnet has fixed poles (1).	1+1+1+1+1
	OR: Electric current produces heat (1½). Furnace reaches very high temperature (1). Melts scrap steel (1). Steel is recycled into useful steel (1½).	1½+1+1+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 3 & 4: **Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Edward Jenner	Smallpox vaccine
2	**(a)** Alexander Fleming	In 1928
3	**(d)** non-communicable diseases	Lifestyle changes
4	**(d)** body, mind and surroundings	Dinacharya and ritucharya
5	**(a)** recover and limit spread	Good practice
6	**(a)** chemicals get used up	Called a dead cell
7	**(b)** copper wire and iron nail	Two different metals
8	**(a)** positive terminal	Positive electrode
9	**(c)** they contain harmful metals	Lead, cadmium, nickel, lithium
10	**(d)** steel manufacturing	Melts and recycles scrap steel

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It avoids illness and suffering (1). It saves cost and prevents spread (1).	1+1
12	Balanced diet and avoiding processed food (1). Regular physical activity and avoiding tobacco (1).	1+1
13	Current flows through it only one way (1). Long lead must go to positive terminal (1).	1+1
14	Helps animals navigate (1). Shields Earth from harmful space particles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stress can disturb sleep (1). Worry can reduce appetite (1). Long stress weakens the body's defence and increases illness risk (1).	1+1+1
16	Use two different metal electrodes in a lemon (1). The juice works as electrolyte (1). A chemical reaction produces small current (1).	1+1+1
17	India is one of the largest vaccine producers (1). It supplies vaccines to many countries (1). Indian companies helped during COVID-19 (1).	1+1+1
	OR: Symptoms are what the child feels, e.g. body ache (1). Signs are fever and rash (1). Consult a doctor, rest and avoid contact (1).	1+1+1
18	Rechargeable batteries wear out with many cycles (1½). Their capacity gradually falls (1½).	1½+1½
	OR: Do not touch bare wires (1). Do not use mains supply for experiments (1). Disconnect circuit when not in use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Both build memory (1). Infection involves actual illness (1). Vaccination is safe and controlled (1). Vaccination protects the community (1). Example: chickenpox (1).	1+1+1+1+1
	OR: Avoids pain (1). Saves cost (1). Prevents spread (1). Vaccines and hygiene (1). Healthy lifestyle reduces NCDs (1).	1+1+1+1+1
20	Use correct wires and plugs (1). Do not touch hot elements (1). Keep batteries away from heat (1). Recycle used batteries (1). Switch off appliances after use (1).	1+1+1+1+1
	OR: Wind coils with different numbers of turns on same nail (1). Use one, then two cells for different current (1). Count paper clips picked up (1). Change one factor at a time (1). More turns or current gives more clips (1).	1+1+1+1+1

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